CYNC ARCHITECTURE INC.

CFD Fire Training Academy - Water Reuse

Scope and Fee Submission

Prepared by CYNC Architecture as Prime Consultant

Project	CFD Fire Training Academy - Water Reuse
SOR No.	24-2020-A01-S06-V06
City Contact	Matthew Scotney, Project Manager, PMO Building Infrastructure, Facilities
Construction Budget	$150,000 Estimated Construction Budget
Submission Deadline	Initial Scope and Fee Submission: July 28, 2026

1. Executive Summary / Project Understanding

CYNC understands that The City of Calgary is seeking a Prime Consultant team to assess and upgrade the existing water-reuse system at the Calgary Fire Department Fire Training Academy, with specific focus on the indoor storage / reuse pond.

As discussed, the project should not begin with a predetermined solution. The first step is to understand the existing system, confirm the intended water-quality performance criteria for fire-training use, evaluate feasible treatment options, and recommend a preferred solution before proceeding into detailed design and construction procurement.

Based on the site review and City background information received, the existing system appears to include an outdoor collection and naturalized filtration / wetland path, an indoor open storage pond within an unheated building, associated pumps, piping, controls, and overflow / stormwater interfaces. The primary concern is maintaining safe and reliable water quality for reuse in training exercises, including consideration of biological water quality and potential exposure pathways such as spray / aerosolization.

The background information and 2026 water testing results support the proposed staged approach. The water quality results reviewed show variable biological conditions over time, including recurring E. coli detections and blue-green algae presence, while Legionella was reported below detection in the samples reviewed. This reinforces the need for Phase 1A assessment and options analysis before committing to a treatment solution.

2. Proposed Consultant Team

Role	Consultant	Scope Intent
Prime Consultant	CYNC Architecture	Overall project leadership, City interface, consultant coordination, Scope and Fee submission, procurement / CA coordination, meeting records, and closeout coordination.
Water Treatment / Process / Mechanical-Process Lead	SweetTech Engineering	Lead technical assessment and treatment strategy, process/mechanical design, options analysis, commissioning strategy, and water-quality performance criteria.
Electrical / Controls / Process Electrical	Catch Engineering, under SweetTech	Electrical review/design for process equipment, power availability, controls, instrumentation, alarms, SCADA/process interfaces, and commissioning support as required.
Wetland / Naturalized System Specialist	To be carried by SweetTech, as required	Review of outdoor wetland / naturalized water path and its relationship to water quality entering the indoor storage pond.

Cost Consultant	BTY	Cost estimates: Class D (up to three options), 60% Class B, and 95% Class A. Life cycle costing excluded unless separately authorized.

3. Existing Background Information Reviewed

The City has provided substantial existing information for review. The background package includes the 2010 Issued for Construction drawings, the 2011 Water Re-Use System O&M Manual, the 2009 Stormwater Management Report, 2026 water sampling results, and the previously proposed aeration scope.

Preliminary observations from the information reviewed include:

- The O&M information indicates that the system was intended to operate as a closed-loop water re-use system, with water collected and re-used for training exercises.
- The 2026 water testing results show variable biological water quality over time. E. coli was detected in the reviewed sample events, with reported results varying by date and sample location.
- Total coliforms were reported in the June 16, 2026 sampling event, and blue-green algae was identified in the reviewed testing results.
- Legionella pneumophila and non-pneumophila were reported below the laboratory reporting limit in the reviewed samples.
- The prior aeration proposal should be treated as background information only. It should not be considered a selected solution until water-quality criteria, treatment objectives, operational requirements, and regulatory considerations are reviewed.

These observations are preliminary and are intended to confirm the need for a disciplined assessment and options analysis. The consultant team will review the data in detail during Phase 1A before recommending a preferred treatment strategy.

4. Proposed Work Plan

CYNC proposes a staged work plan that aligns with the City's SOR while clearly distinguishing Basic Services, Additional Services, and Specialty Consultant Services.

Phase 1A – Technical Business Case / Scope Confirmation

This stage is the critical front-end portion of the project and should be identified primarily as *Additional and Specialty Consultant Services*. The objective is to understand the existing system, define the water-quality problem, establish performance criteria, and recommend a preferred path forward for further assessment, engineering, and design.

- Attend project kickoff meeting with The City.
- Review provided background information, including 2010 IFC drawings, the 2011 O&M Manual, 2009 stormwater management report, 2026 water-quality test results, prior aeration information, operational procedures, and City regulatory background information.
- Complete site and operational review of the indoor pond, outdoor pond / wetland / river system, pumps, piping, controls, overflow connections, and maintenance access.
- Review current manual chlorine dosing approach and any previous or proposed aeration work.
- Review biological, physical, and chemical water-quality concerns, including potential exposure pathways related to spray / aerosolization during fire-training use.
- Confirm water-quality objectives appropriate for outdoor fire-training use, noting that diving / full immersion use is excluded unless otherwise directed by The City.
- Identify applicable regulatory, approval, and permitting considerations, including confirmation of whether specialized regulatory coordination is required.

- Review existing 2026 water testing results, identify whether additional water testing is required, and recommend initial testing parameters if needed.
- Evaluate treatment options, which may include automated chlorination, aeration, UV disinfection, filtration / circulation improvements, pond cleaning / sediment management, operational controls, and other recommended measures.
- Prepare a technical memo summarizing options, capital cost considerations, operating cost considerations, advantages, disadvantages, risks, and recommended path forward and, where possible, preferred options for further development.
- Present the technical memo to The City and respond to review comments.
- Options-stage pricing is order-of-magnitude, with more refined estimates at 60% and 95%.

Phase 1B – Detailed Assessment, Engineering, Design and Tender Documents — to be confirmed following Phase 1A

Following The City acceptance of the preferred option, the team will develop the selected solution through design submissions and procurement documentation.

- Prepare 30% design submission for City / TAS review.
- Prepare 60% design submission and updated cost estimate.
- Prepare 95% design submission and updated cost estimate.
- Prepare drawings and specifications for the selected treatment upgrades.
- Develop process/mechanical design, electrical/process-controls design, and equipment requirements as required by the selected option.
- Coordinate permit and approval applications, if required.
- Prepare construction procurement documents suitable for the City procurement process.
- Chair regular design meetings and prepare meeting minutes and action items.

Phase 2 - Construction Procurement, Contract Administration and Commissioning

Phase 2 should be shown separately and identified as conditional on written City approval of construction funding based on the 95% construction cost estimate.

- Support City construction procurement process, including bidder questions, RFIs, and addenda.
- Assist with bid review, evaluation, and recommendation for award.
- Issue coordinated IFC documents, incorporating addenda as required.
- Chair construction meetings and prepare meeting records.
- Review shop drawings, materials, RFIs, Site Instructions, proposed changes, and other typical CA documentation.
- Complete site reviews and issue site review reports.
- Attend equipment startups and provide commissioning review / reporting for installed treatment equipment.
- Confirm that water-quality performance criteria established during Phase 1 have been achieved, based on agreed testing and verification process.
- Review progress claims and provide payment certification recommendations.
- Conduct deficiency reviews, closeout review, and warranty follow-up as required.

5. Fee Framework

The Category A Basic Services fee has been shown separately based on 13.5% of the City’s estimated $150,000 construction budget. Additional and Specialty Consultant Services required to complete the front-end assessment, treatment/process review, environmental/wetland review, cost consulting, and testing coordination are identified separately.

Fee Item	Category	Consultant(s)	Fee
Basic Services — 13.5% of $150,000 construction budget	Basic Services	CYNC / Design Team	$20,500
SweetTech Phase 1A technical business case / desktop review	Additional Services	SweetTech	$12,000
Water Treatment / Process Options Analysis	Specialty Consultant Services	SweetTech	TBD
Wetland / Naturalized System Review	Specialty Consultant Services	SweetTech subconsultant	TBD
Electrical / Controls / SCADA Review	Specialty Consultant Services	Catch / SweetTech	TBD
Cost Consulting	Specialty Consultant Services	BTY	Future/As required
Water Testing / Lab Analysis	Allowance / Reimbursable	SweetTech / Lab	TBD
Phase 1B and Phase 2 Construction Procurement, Contract Administration and Commissioning	Conditional Phase	CYNC / SweetTech / Catch / BTY	To be confirmed following Phase 1A
Phase 1A Subtotal:			$32,500
Disbursements:			$1,625
Phase 1A Fixed Fee / PO Total:			**$34,125 + GST**

Fees are exclusive of GST and reimbursable expenses unless noted otherwise.

Depending on the City's preferred authorization approach, Phase 1A Assessment and Options Analysis may be authorized separately from Phase 1B Design and Procurement Documents.

6. Additional / Specialty Service Rationale

The following items should be explicitly identified as Additional and/or Specialty Consultant Services above the Category A Basic Services fee:

- Pre-design phase work required to define the problem and potential treatment strategies before design proceeds.
- Existing facility and existing system surveys, including review of indoor pond, outdoor wetland / river, pumps, piping, controls, dosing, and overflow interfaces.
- Water-treatment / process engineering and design, including assessment of chlorination, aeration, UV disinfection, filtration, circulation, and operational controls.
- Environmental and wetland specialist input related to the naturalized filtration path and water-quality implications.
- Regulatory and approval review beyond normal building permit coordination.
- Cost consulting for options analysis, capital cost review, operating cost considerations, and 60% / 95% estimates.
- Water sampling and laboratory analysis, if required, as a separate allowance or reimbursable item.

7. Key Assumptions and Exclusions

- Construction budget is assumed to be $150,000, as identified in the SOR.
- Phase 2 services are conditional on written City approval of construction funding.
- Water testing and laboratory analysis are assumed to be carried as a separate allowance or reimbursable expense. The consultant team will recommend testing parameters, frequency, and interpretation requirements after reviewing existing water-quality data. The City may elect to retain the laboratory directly, provided testing can be completed within the project schedule.
- The preferred treatment option has not yet been selected; detailed design scope may need to be confirmed following completion of Phase 1A options analysis.

- The scope includes review of the exterior pond / wetland / naturalized system as it relates to indoor pond water quality. Full redesign of the exterior system is excluded unless specifically authorized as an additional service.
- Detailed regulatory applications beyond those identified during Phase 1A may require additional authorization.
- Hazardous materials testing, confined space entry services, and specialized health and safety studies are excluded unless specifically identified and authorized.
- Long-term operating procedures and testing protocols will be developed after the preferred treatment strategy is selected, unless directed otherwise by The City.

8. Approach

CYNC's approach is to first define the problem, confirm performance criteria, and identify the most appropriate treatment strategy before proceeding into detailed design. This staged approach is intended to protect the City's budget, avoid premature commitment to an unsuitable solution, and provide a clear path to construction, commissioning, and safe long-term operation.

9. Key Decision Points

Decision Point	Purpose
Completion of Phase 1A technical memo	Confirm preferred treatment strategy
Completion of 60% design estimate	Confirm scope and budget alignment
Completion of 95% design estimate	Confirm construction funding and proceed/no-proceed for Phase 2
Post-construction commissioning	Confirm water-quality performance criteria are achieved

10. Background Information Provided / To Be Confirmed During Phase 1A

To be confirmed during Phase 1A: current operating records and current chlorine dosing procedures, if separate from the O&M information.
To be confirmed during Phase 1A: City regulatory / Water Act review notes and any current water-quality thresholds being used by The City.



SCHEDULE 3 -
SAMPLE SCOPE AND FEE SCHEDULE

The scope and fee will be set out in the attached form as indicated in Section 6.3.4 of the RFP.

Scope and Fee Schedule	**24-2020-A01-S06-V06**
Project: CFD-FTA Water Reuse	
Consultant Name: CYNC Architecture Inc.	
The intent of this form is to finalize the scope and fee negotiation from the RFx process. This form is expected to be an iterative process between the Consultant and the City Representative, with the document not complete until fully signed-off in Part D below. The Scope and Fee Schedule is only legally binding after the execution of the Agreement.	

PART A: SCOPE OF WORK

Prime Consultant shall perform investigation, assessment, design, procurement support, and construction administration services for the Project, with a focus on implementing measures to improve and maintain the water quality of the indoor storage pond. The Prime Consultant shall be responsible for delivering a complete and coordinated design solution that can be constructed and operated to meet the Project's performance requirements. The Prime Consultant shall retain the necessary subconsultants as necessary, including, but not limited to the following:

- Civil Engineering
- Mechanical Engineering
- Electrical Engineering
- Environmental Engineering
- Cost Consulting
  Note: Subconsultants shall have experience with water and/or wastewater treatment, and reference projects shall be provided for review.

The Prime Consultant will be responsible for coordinating with The City and any other consultants retained by The City for the duration of the project.

Key tasks and deliverables required, but not limited to, are:

Phase 1 – Evaluation, Options Analysis, and Design

1. Attend a project start-up meeting, hosted by The City.
2. Complete site assessment(s) and an in-depth review of the existing water reuse system, with particular focus on the indoor storage facility, associated piping, pumping, treatment infrastructure, and interfaces with the overall training-water and stormwater system. The overall system includes the training area collection, raw water pond, engineering wetland, indoor storage and reuse pond, stormwater/overflow pond, and associated infrastructure.
3. Review available background information, existing drawings, reports, operational information, available testing information (most recent water tests are from June 2026, with future testing planned), and any information related to the previous or proposed improvements. This shall include review of the manual

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chlorine dosing procedures, and the previously procured small aeration system that was procured, but postponed.

4. Meet with The City to confirm project requirements, operational needs, performance objectives, maintenance considerations, and any health and safety concerns related to use of reused water in training exercises.
5. Review current system concerns including, but not limited to, biological water quality, potential staff exposure pathways such as aerosolization, effectiveness and long-term operability of the current chlorine dosing approach, and suitability of water quality for use in the outdoor training area.
6. Identify applicable codes, regulations, approvals, and permitting requirements relevant to the proposed upgrades and confirm whether any specialized regulatory coordination will be required. The City has conducted preliminary reviews related to the Water Act and applicable regulations. The info will be provided for background; however, the Prime Consultant shall independently verify all regulatory requirements and conclusions.
7. Make recommendations/provide requirements on the specific physical, chemical, and biological water quality conditions that need to be achieved through the improvements to the system, and the types, and schedule of tests that should be completed to track long term water quality. The water quality requirements shall be appropriate for use in outdoor training activities and consider potential exposure pathways (e.g., spray/aerosolization), but not diving, or full immersion within the pond.
8. Prepare a technical memo outlining the potential options and recommendations for improving and maintaining water quality. Include design costs, capital costs, operating costs, advantages/disadvantages, risks, and a recommendation on the preferred option(s). Provide a presentation on this memo, and respond to questions from The City. Potential options to review can include, but are not limited to:
    a. Enhanced or automated chlorination systems;
    b. Aeration systems;
    c. Ultraviolet (UV) disinfection; and
    d. Other treatments, controls, operational, or infrastructure improvements recommended by the Prime Consultant.
9. As part of the options analysis, assess constructability, operational impacts, maintenance implications, implementation risks, and compatibility with the existing system, including consideration of overflow/stormwater impacts where relevant.
10. Review the previously proposed aeration approach and advise whether it is suitable for this site and project, and whether it should proceed as a standalone implementation or be incorporated into the broader construction package.
11. Provide recommendations on a preferred option(s) for review and acceptance by The City.
12. Provide a project schedule for design deliverables and key review/decision milestones.
13. Submit 30%, 60%, and 95% contract document packages for review by The City. Allow for a 3 week review period with each submission, for The City's Technical & Advisory Services (TAS). Include updated cost estimates at the 60% and 95% complete submissions.
14. Prepare construction procurement documents (drawings and specifications) for the selected option(s).
15. Prepare all required permit and approval applications and supporting documentation, as applicable, and coordinate with the relevant authorities and The City through the review process. The City shall pay the application fees for the relevant permits.
16. Chair regular design meetings, including preparing and distributing meeting minutes and action items.

Phase 2 – Construction Procurement, Contract Award, and Contract Administration

Note: proceeding into Phase 2 is conditional upon written confirmation of construction funding approval, based on the Phase 1, 95% complete cost estimate from the design.

1. Support The City with the construction procurement process, including responding to technical questions, reviewing Requests for Information (RFIs), and preparing addenda, as required
2. Assist with tender/bid document review, evaluation of submissions, and provide a recommendation for award.

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3. Provide full coordinated, Issued-For-Construction documents, including final drawing and specifications updates, and any addendum revisions, as applicable.
4. Chair regular/bi-weekly construction meetings throughout the duration of construction.
5. Provide contract administration services during construction, including review of shop drawings, materials, responding to RFIs, issuing Site Instructions, processing changes, and other typical contract administration documentation, as applicable to the project. Make use of City of Calgary forms where applicable.
6. Carry out site reviews during construction, and issue site review reports following site visits.
7. Coordinate with all required subconsultants through construction.
8. Attend equipment start-ups, and provide commissioning services and reports for any installed equipment. Include confirmation that water quality targets (established in Phase 1) are achieved.
9. Review progress claims and provide payment certification recommendations on a monthly basis. Conduct reviews within the standard timelines required by The City, and the Prompt Payment and Construction Lien Act.
10. Conduct deficiency reviews when required, provide coordinated deficiency lists, and attend substantial completion and final completion inspections as applicable. Review contractor-provided closeout documentation, including Operations & Maintenance Manual, commissioning/start-up information, and as-built record documentation, as applicable to the project scope.
11. Prepare final record drawings (CAD files) to document as-built conditions
12. Provide post-construction support during the warranty period, and/or for deficiency follow-ups.

PART B: PROJECT SCHEDULE

Activity	Timeline
Target Start of Work	**August 2026**
Phase 1A – Technical Business Case | Scope Confirmation	**August 2026 - September 2026 or Approximately 4–6 weeks from authorization**
Phase 1B - Detailed Assessment, Engineering, Design and Tender Documents	**To be confirmed following Phase 1A**
Phase 2 - Construction Procurement, Contract Administration and Commissioning	**To be confirmed following Phase 1B and construction funding approval**
Target End Date	**To be confirmed following Phase 1A / Phase 1B scope confirmation**

PART C: KEY PERSONNEL

Title/Role	Name, Firm
Project Lead	*Gordon Murray, CYNC ARCHITECTURE INC.*
Architect of Record	*Gordon Murray, CYNC ARCHITECTURE INC.*
Structural Lead/Engineer of Record	*Not Applicable*
Mechanical Lead/Engineer of Record	*Shaun Robertson, SweetTech Engineering*

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Electrical Lead/Engineer of Record	*Scott Pederson, P.Eng., Catch Engineering, under SweetTech*
Civil Lead/Engineer of Record	*Austyn Schutta, P.Eng., SweetTech Engineering / wetland specialist as applicable*

PART D: FEE BREAKDOWN

Deliverable/Phase	Base Fee	Extra Services Fee	Total
Phase 1A – Technical Business Case / Scope Confirmation			
CYNC Prime Consultant coordination, City interface, submission coordination, consultant management, review / issue of technical memo	$20,500		**$20,500**
SweetTech technical business case / desktop review / technical memo support, including Catch and wetland specialist input		**$12,000**	**$12,000**
YJE building M&E / electrical interface allowance, carried within CYNC fee	Included		**Included**
BTY Cost Consulting	Future / as required	Future / as required	**TBD**
Phase 1A Sub-Total	$20,500	$12,000	**$32,500 (+GST)**
Phase 1B – Detailed Assessment, Engineering, Design and Tender Documents	To be confirmed following Phase 1A	To be confirmed following Phase 1A	**TBD**
Phase 2 – Construction Procurement, Contract Administration and Commissioning	To be confirmed following Phase 1B / funding approval	To be confirmed following Phase 1B / funding approval	**TBD**

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Disbursements	$1,025	$600	**$1,625**
Fixed Fee / PO Total for Phase 1A	$21,525	$12,600	**$34,125 + GST**

SweetTech's Phase 1A technical business case / desktop review is identified as an Extra Services Fee because it relates to specialized water-treatment, process, wetland, and electrical/controls input required to define the appropriate path forward.

PART E: ACKNOWLEGEMENT AND SIGN-OFF

After the City and the Consultant have agreed to the Scope and Fee Schedule as set out above, the Consultant's Representative will sign below to indicate their acknowledgement. This document will be submitted with The Agreement for signatures.

Agreed to on this date

For the Consultant (Print Legal Name)

GORDON MURRAY

Per: _____

Authorized Signatory

GORDON MURRAY

Print name

PRINCIPAL

Print title

DRAFT

For internal use by The City:

The Business Unit approves the content of this document.

Print DeptID Name _____

DeptID Signature _____

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CERTIFICATE OF INSURANCE

CERTIFICATE HOLDER:
City of Calgary
P.O. Box 2100, Stn. M
Calgary, AB T2P 2M5

Dated: July 24, 2026

RE: SOR 24-2020-A01-S06-V06 - CFD Fire Training Academy – Water Reuse

NAMED INSURED:
CYNC Architecture Inc.
Suite 200 - 337 17 Ave SW
Calgary, AB T2S 0A5

BROKER:
Navacord Insurance Services Alberta Inc.
Suite 100, 10120 - 103 Avenue NW
Edmonton, AB T5J 3R6 P: (780) 483-4544

This certificate is issued as a matter of information only and confers no rights upon the certificate holder. This certificate does not amend, extend, or alter the coverage afforded by the policies below. The insurance afforded is subject to the terms, conditions, and exclusions of the applicable policy.

COMPANIES AFFORDING COVERAGE:

Company Letter "A" Northbridge General Insurance Corporation
Company Letter "B" XL Specialty Insurance Company

CO LTR	TYPE OF INSURANCE	POLICY NUMBER	POLICY EFFECTIVE DATE	POLICY EXPIRY DATE	LIMIT	
GENERAL LIABILITY						
A	Commercial General Liability including: - Broad Form Property Damage - Personal & Advertising Injury - Cross Liability/Severability of Interest - Contingent Employers Liability - Incidental Malpractice Liability	P04028366	March 11, 2026	March 11, 2027	$2,000,000 $2,000,000 $2,000,000	Per Occurrence General Aggregate Aggregate Products & Completed Operations
A	Employers Liability	P04028366	March 11, 2026	March 11, 2027	$2,000,000	Per Occurrence
A	Employee Benefits Liability	P04028366	March 11, 2026	March 11, 2027	$1,000,000	Per Occurrence
A	Tenants Legal Liability	P04028366	March 11, 2026	March 11, 2027	$500,000	Per Occurrence
A	Non-Owned Automobile	P04028366	March 11, 2026	March 11, 2027	$2,000,000	Per Occurrence
A	SEF 94 - Legal Liability for Damage to Hired Automobiles	P04028366	March 11, 2026	March 11, 2027	$75,000	Per Occurrence
ERRORS & OMISSIONS LIABILITY						
B	Professional Liability - Design Professionals (A&E) E&O	DPX 9972050	March 4, 2026	March 4, 2027	$2,000,000 $2,000,000	Each Claim Aggregate

CANCELLATION
Should the Commercial General Liability policy be cancelled before the expiration date thereof, the issuing company will endeavor to mail ***nil*** days written notice to the certificate holder named above, but failure to mail such notice shall impose no obligation or liability of any kind upon the company, its agents or representative.

Navacord Insurance Services Alberta Inc.
Per: John Bacon

NAVACORD®

Workers' Compensation Board – Alberta

9912 – 107 Street
PO Box 2415
Edmonton AB T5J 2S5

Email: ebusiness.support@wcb.ab.ca
Tel: (780) 498-3999 (1-866-922-9221)
Fax: (780) 498-7999
WCB website: www.wcb.ab.ca

July 24, 2026

Reference Number: 11427

MATHEW SCOTNEY
CITY OF CALGARY
ATTENTION: WCB COORDINATOR
PO BOX 2100, STN M, MC 8020
CALGARY AB T2P 2M5

Dear Sir or Madam:

Re: CYNC ARCHITECTURE INC.
204-337 17 AVE SW
CALGARY AB T2S 0A5

FTA - WATER REUSE - RFSO 24-20

The above named subcontractor has an account with WCB-Alberta in the following industry(ies):

account	trade names(s)/industry	effective date	coverage
6369832	ARCHITECTS, DRAFTING & DESIGN	Feb 16, 2011	worker coverage personal coverage for: GORDON C MURRAY

Thank you for checking into the status of this contractor or subcontractor. Under Section 126 of the Workers' Compensation Act, you are responsible for obtaining a clearance on your contractor or subcontractor, in order to release you from any liability for unpaid WCB premiums owed by them. Please ensure clearance has been issued in the correct name and that there is coverage in the industry(ies) for which work was performed.

Please accept this letter as a clearance for work completed between the effective date of the account and the date of this letter. For this account, you are cleared of any liability under Section 126 of the Workers' Compensation Act up to the date of this letter. Any holdback may be released for contracts completed, and/or for work completed to the date of this letter. For an account that shows closed under the effective date, the clearance is only valid for work completed up to the close date. If work has not started, obtain a clearance prior to releasing final payment.

Please note, if any directors of the corporation are injured at work, you are protected from lawsuit if they have personal coverage. If they do not have personal coverage, you may not be protected in the case of a workplace injury.

If your contractor or subcontractor is performing work outside Alberta, contact the WCB in that jurisdiction to determine your clearance and any other WCB requirements.

Any alteration of this document is strictly prohibited.

Yours truly,

eBusiness Support Team (16712531)

SIGN UP FOR ONLINE SERVI... GO TO MY WCB-AB...



GORDON MURRAY AAA, AIBC, SAA, FRAIC

Principal | Architect | Founder | CYNC Architecture Inc. | 403.269.7782 | gordon.murray@cync.ca

PROPOSED ROLE

- Principal-in-Charge / Project Director
- Senior client contact
- Prime consultant leadership
- QA review and final review

CLIENT RELEVANCE

- 30+ years of architectural leadership
- Industrial, modular, utility, healthcare, public-sector and northern work
- Existing building review and consultant coordination across Western and Northern Canada

CORE STRENGTHS

- Prime consultant leadership
- Industrial and modular planning
- Existing building review
- Code and accessibility review
- Building envelope coordination
- Consultant coordination
- Stakeholder and client communication
- Quality assurance and senior review

PROFESSIONAL STANDING

- Licensed Architect: AB, BC, SK, NU, NWT, Yukon
- President, Alberta Association of Architects, 2020-2021
- Fellow, Royal Architectural Institute of Canada
- Public-sector pre-qualifications: City of Calgary, Rocky View Schools, Alberta Infrastructure, Government of Nunavut

EDUCATION

- Master of Architecture, University of Calgary
- Bachelor of Fine Arts, University of Calgary
- Emily Carr College of Art, Vancouver

EXECUTIVE PROFILE

Founder and Principal of CYNC Architecture Inc. with more than three decades of experience leading architectural practice, client strategy, consultant coordination, technical review, quality assurance, and project delivery. Gordon has recent direct experience with modular classroom programs, technically coordinated industrial facilities, utility infrastructure, healthcare upgrades, northern maintenance buildings, and food-production / recycling facility assignments. His work emphasizes practical planning, clear communication, code-aware review, durable detailing, and coordination of consultant input in active operational environments.

PROFESSIONAL EXPERIENCE

Principal / Founder | CYNC Architecture Inc. | 2019-Present - Leads architectural practice, client strategy, consultant coordination, technical review, quality assurance, and project delivery for public, institutional, commercial, industrial, modular, healthcare, utility, and northern projects.

Principal | Iredale Architecture Calgary | 2017-2019 - Provided senior architectural leadership and client coordination for institutional, public-sector, commercial, and northern project work.

Principal / Founder | Metro Architecture / Duel Architecture Inc. | 2006-2016 - Founded and led architectural practice delivering commercial, institutional, public-sector, residential, and mixed-use projects.

Earlier Roles | Riddell Kurczaba; DMC Resources; Cohos Evamy; Poon McKenzie; The LeBlond Partnership | 1992-2006 - Built broad technical and delivery experience across design, documentation, consultant coordination, and construction-phase services.

SELECTED INDUSTRIAL / MODULAR / TECHNICAL PROJECT EXPERIENCE

Calgary Board of Education - Modular Classroom Programs | Calgary, AB | 2026-Present - Prime consultant services for multiple CBE modular classroom assignments, including existing condition review, site and building coordination, public-sector communication, accessibility/code awareness, consultant coordination, documentation, and work within active school environments. Relevant sites include Belfast, Colonel Irvine, Dr. E.P. Scarlett, Ernest Manning, Grant MacEwan, Hugh A. Bennett, Joane Cardinal-Schubert, Kenneth D. Taylor, Lester B. Pearson, Mahogany, North Trail, Olympic Heights, John G. Diefenbaker, Mount Royal, Centennial, Annie Foote, Sibylla Kiddle, David Thompson, Marlborough, and Windsor Park.

Cambridge Bay Power Plant | Nunavut | 2025 - Architectural services scope for 350,000 SF building, architectural / building enclosure design, gap matrix development, coordination with engineering consultants for fire/life-safety/code consulting, IFT coordination and defined RPR candidate responsibilities for selected architectural and building enclosure items.

Synapse Data Centre | Olds, AB - Architectural design and coordination for a technically complex data-centre facility involving equipment zones, service access, envelope coordination, life-safety considerations, consultant coordination, and coordination of building systems within an operationally focused facility.

Teknion Solutions Project | Alberta - Architectural services for an industrial / technical facility assignment, including existing condition review, code-aware planning, consultant coordination, documentation, and support for practical implementation within a working operational context.

ENMAX Utility Facility Assignments | Calgary / Alberta - Architectural services and code-aware coordination for utility and substation-related work, including existing facility renovation, operational continuity, equipment access, durable materials, consultant coordination, and work within constrained technical environments.

AHS Didsbury District Health Services - Electrical Closet / Distribution Upgrades | Didsbury, AB - Architectural support and coordination for electrical closet and infrastructure renewal work in an existing healthcare facility, with attention to operational continuity, existing conditions, code coordination, consultant input, and phased implementation requirements.

Government of Nunavut - Airport Maintenance Buildings | Nunavut, Canada - Architectural design and technical coordination for heavy equipment maintenance facilities requiring durable planning, operational review, equipment coordination, accessibility/code awareness, and northern climate considerations.

SEH / Inland Steel Recycling Facility | Lloydminster, SK - Architectural support for a recycling / industrial facility assignment, including practical planning, consultant coordination, code-aware review, operational considerations, and support for documentation suited to a working industrial site.

Shearer's Potato Chip Facility | Alberta - Architectural services for a food-production facility assignment requiring practical coordination of industrial operations, durable finishes, code-aware planning, consultant coordination, and support for implementation in an active production-oriented environment.

RELEVANCE TO INDUSTRIAL AND MODULAR ASSIGNMENTS

Gordon provides senior leadership for assignments that require careful coordination between client operations, building code requirements, existing conditions, technical consultant input, and practical construction constraints. His recent industrial and modular work supports projects where schedule awareness, clear review comments, durable material decisions, and disciplined consultant coordination are central to successful delivery.

CYNC ARCHITECTURE INC.
337 17 AVE SW
Calgary Alberta
T2S 0A5
403.269.7782
cync.ca

SWEETTECH
ENGINEERING CONSULTANTS

July 27, 2026 FILE: P26.2313.003

Gordon Murray, AAA, AIBC, SAA, FRAIC
CYNC Architecture Inc.
204, 337 17 Avenue SW
Calgary, Alberta T2S 0A5
Via email: Gordon.Murray@cync.ca

RE CFD Fire Training Academy – Water Reuse, SOR 24-2020-A01-S06-V06

Dear Gordon,

Further to your request, SweetTech Engineering Consultants (SweetTech) is pleased to provide this scope of work for technical consulting services associated with the Calgary Fire Department Fire Training Academy water-reuse project. This letter is intended to accompany SweetTech's fee estimate and to form the basis of our subconsultant services to CYNC Architecture Inc. (Prime Consultant).

SweetTech has recently completed (and is completing) a number of multidisciplinary projects involving natural and mechanical water treatment and is well positioned to help CYNC and the City as subject matter experts in project management and assessment techniques of complex natural water treatment systems.

SweetTech understands that the existing water-reuse system requires a practical, budget-driven review focused on improving and maintaining water quality in the indoor storage and reuse pond. The system must support outdoor hose and water-training exercises, including potential exposure through spray and aerosolization, while remaining compatible with the site's existing ponds, engineered wetland, stormwater system, piping, pumps, controls, and treatment infrastructure. Detailed assessments, design, and engineering will be required to achieve this outcome, which SweetTech will help support the technical scoping of.

ROLES AND RESPONSIBILITIES

The Prime Consultant will retain the prime agreement with The City of Calgary, act as the formal City contact, manage the overall project agreement, and retain contractual responsibility for the complete consultant assignment. SweetTech will serve as the Prime Consultant's technical subconsultant and perform or co-ordinate the technical requirements described below.

SweetTech's team includes a Wetland and Stormwater Specialist subconsultant to inform specifics regarding the existing wetland and an Electrical Engineering Subconsultant to inform specifics regarding the existing electrical and automation. SweetTech will co-ordinate the civil, mechanical/process, electrical, environmental, regulatory, and cost-consulting inputs required to inform the below Phase 1 Scope. The final discipline assignments and subconsultant relationships will be confirmed with the Prime Consultant before authorization.

Unless otherwise agreed, the Prime Consultant will:

- manage the prime contract, invoicing, commercial correspondence, and formal submissions to The City;
- lead executive-level client communication and project governance;
- provide available background information and obtain City decisions, approvals, and authorizations;
- confirm submission formatting, document-control, insurance, safety, and contractual requirements; and
- review and issue SweetTech's technical work as part of the Prime Consultant's coordinated deliverables.

PHASE 1 SCOPE

SweetTech Engineering will support CYNC and the City to develop a Plan to evaluate and design a solution to the existing System at a later date. Detailed engineering and design of solutions are outside of this scope, but these activities can be performed by SweetTech at a later date.

SweetTech will aid CYNC and the City in evaluating the project scope, providing options in how to scope the future work, and aid in design of a project management framework for project delivery. We see this exercise as development of a technical Business Case and Scope of Work, with an emphasis on outlining the technical requirements and path forward needed to support the future Work.

Our scope, as defined by CYNC, is suggested to include:

- Initial meetings with the City of Calgary;
- Emails, phone calls, and communication with CYNC;
- Desktop-only review of existing information;
- A brief memorandum that will include:
    - Situational Summary
    - Needs Identification
    - Desired Outcomes
    - Constraints
    - Options for Project Execution
    - Recommendation for Path Forward incl. Schedule and Estimated Fees for Technical/Professional Services required to support the execution of the future tasks: technical studies, engineering, review and assessment, design, drafting, and Tender document development for Construction Procurement.

PROPOSED PROJECT TEAM

The above Work will be supported by the following staff:

SHAUN ROBERTSON, P.Eng.

PROJECT LEAD / MECHANICAL AND PROCESS ENGINEERING LEAD

Credentials: P.Eng.; Practice Lead, Mechanical and Process Engineering; 20+ years of mechanical, process, pumping, piping, hydraulic analysis, and field focused project delivery experience in fluid process and treatment.

Shaun Robertson will serve as SweetTech's project lead and mechanical/process engineering lead for this planning-level assignment. As SweetTech's Practice Lead, Mechanical and Process Engineering, he brings more than 20 years of practical experience with pumping systems, hydraulic analysis, process piping, equipment sizing, production facilities, and field-focused project delivery. Shaun will direct SweetTech's desktop review, coordinate input from Austyn Schutta, and keep the assessment focused on the project's practical needs, constraints, desired outcomes, and execution options.

Shaun's experience is well suited to an early-stage assessment where the immediate need is to understand the available information, identify gaps and constraints, and develop a practical path forward before committing to detailed engineering. He will apply a mechanical and process perspective to the situational review and needs assessment, considering how existing or proposed systems, equipment, piping, operations, maintainability, constructability, and future implementation requirements could influence the available options. His input will support a concise memorandum that gives CYNC and The City a clear basis for the next project decision.

As project lead, Shaun will participate in the initial meetings, maintain direct communication with CYNC, coordinate SweetTech's desktop review, and lead preparation of the brief memorandum. He will work closely with Austyn to consolidate the situational summary, needs, desired outcomes, constraints, execution options, and recommended path forward, including a planning-level schedule and cost basis. Shaun's delivery style reflects SweetTech's small-firm advantage: direct access to the senior professionals completing the work, organic and immediate communication, and frank advice where scope decisions may affect cost, schedule, operations, or risk.

AUSTYN SCHUTTA P.Eng., CPESC, CESSWI

SR. CIVIL & HYDROTECHNICAL ENGINEER

Credentials: P.Eng. of 14+ years Coordinating Professional and Engineer of Record on numerous multi-disciplinary projects involving development and execution of technical Scopes of Work involving water management, systems process, regulatory, construction Contracts and detailed Public Tenders.

Given the variety of experience in his career working for contractors, owners, and consultants across three continents, Austyn understands well the roles of each party in a project setting and is highly sought after as a project manager, civil engineer, and contract administrator for environmentally sensitive and nuanced work. While working for contractors, he has been responsible for project management, quality control, field engineering, quantity and cost tracking, scheduling, safety, and document control on multi-disciplinary civil projects. He has developed project control systems and scheduling and estimating programs for companies, municipalities, and large project teams. Austyn is able to communicate information effectively between all parties and has a strong field-level and technical background in engineering principles and theory, constructability, regulatory conditions, contract administration, and project execution. He is able to apply this experience at every stage of a project to proactively approach problems and develop practical solutions. His successful project management experience, spanning over 15 years, ranges from large (~$995M) to small (~$50K) projects, understanding the needs and demands different projects require. He has successfully managed large and small teams of professionals and staff to achieve successful project outcomes.

As a Civil Engineer, Austyn's practice complements the scoping and advisory work described above. He has led multidisciplinary planning and assessment assignments involving new facilities, refurbishments, replacements, site servicing, drainage, earthworks, environmental constraints, and regulatory considerations. This experience positions him to review the available information, identify project needs and constraints, co-ordinate specialist input, evaluate practical execution options, and develop a recommended path forward with an appropriate schedule and planning-level cost basis. Austyn is a Professional Engineer registered with APEGA and EGBC, a Certified Professional in Erosion and Sediment Control (CPESC), a Certified Erosion, Sediment and Stormwater Inspector (CESSWI), and a member of the Association for the Advancement of Cost Engineering (AACE) and the Canadian Water Resources Association (CWRA).

SCOTT PEDERSON, P.Eng.

ELECTRICAL ENGINEERING LEAD - CATCH ENGINEERING

Credentials: P.Eng. of 20+ years of electrical engineering experience including industrial electrical systems, utility interconnections, protection and control, medium - and high-voltage equipment, and electrical safety/code review.

Scott Pederson will serve as the electrical engineering lead from Catch Engineering. Scott is a Professional Electrical Engineer with more than 20 years of experience in the design, coordination, and review of complex industrial electrical systems. His background includes technical governance, engineering assurance, design certification, medium and high voltage equipment, protection and control schematics, utility interconnections, and electrical safety/code considerations. He is registered with APEGA and EGBC and brings a senior review mindset suited to a public water infrastructure project where reliability and maintainability matter. Scott will provide input into the electrical and automation systems portions of this project.

CORY ALBERS P.Eng., M.Sc.

SUBJECT MATTER EXPERT– WETLANDS & HYDROLOGY ENGINEER

Credentials: P.Eng., M.Sc.; more than 20 years of experience in wetland hydraulic systems, forensic hydraulics, ecological design, and naturalized water treatment for stormwater, wastewater, and complex water infrastructure. Co-inventor of the WaterKidney® and Nautilus Pond® systems and recipient of the 2024 Water Resources Award of Excellence.

Cory Albers, P.Eng., M.Sc., is Co-Owner of Source2Source Inc. in Calgary, Alberta. With over 15 years of experience in water resources engineering, he specializes in advanced hydraulic systems design and analysis for both open channel and closed conduit applications.

Cory delivers technical depth in fluid flow modeling while integrating broader ecological and physical system interactions. He excels at collaborating with multidisciplinary teams, including biologists, landscape architects, regulators, contractors, and clients across public and private sectors.

His expertise has positioned him as a lead hydraulic analyst on high-profile water and wastewater treatment projects in the Calgary region. Cory is also a key contributor to innovative stormwater management solutions, such as the Stormwater Kidney™ retrofit projects that address urbanization, flood control, water quality, and conservation challenges.

FEE BASIS

In our experience, Assessment, Engineering, Design, and Public Tender Procurement for a brownfield system of this complexity involving mechanical, electrical, and environmental process will require budgetary fees for professional services in the range of $150,000 - $300,000. As such, the $20,000 overall budget to develop a project management strategy and framework seems appropriate.

Understanding that CYNC will be the Prime Consultant, they will be responsible for directing SweetTech with regards to scope and budget.

We recommend an initial budget of $12,000 + GST for SweetTech to support the above Work working on a Time and Materials basis.

Staff	Hourly Rate ($/h)	Estimated Hours
Shaun Robertson, P.Eng.	225	30
Austyn Schutta, P.Eng., CPESC, CESSWI	185	16
Scott Pederson, P.Eng.	225	4
Cory Albers, P.Eng., M.Sc.	345	4

SCHEDULE

SweetTech is prepared to support the target Phase 1 period of August 2026 through February 2027 subject to timely authorization, receipt of background information, specialist availability, City review periods, and decisions.

We believe there is opportunity to accelerate this schedule as we believe our portion of the work can be completed in about 4 to 6 weeks total with consideration for existing workload and vacation schedule.

ASSUMPTIONS AND EXCLUSIONS

SweetTech's scope and fee are based on the following assumptions and exclusions:

- Services will be completed on a time-and-materials basis to an initial budget of $12,000 plus GST. SweetTech will notify CYNC before exceeding the authorized budget.
- The assignment is a planning-level, desktop-only review intended to define project needs, constraints, execution options, and a recommended path forward. It is not detailed engineering or design.
- CYNC will act as Prime Consultant, direct SweetTech's work, manage the prime agreement, and issue coordinated communications and deliverables to The City.
- CYNC and The City will provide available drawings, reports, operating information, water-quality data, previous studies, and other relevant records in a timely manner. SweetTech may rely on this information without independent verification.

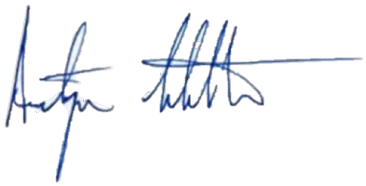
- The fee includes initial meetings, reasonable email and telephone communication, desktop review, specialist input within the hours shown in the fee table, and preparation of one brief memorandum.
- The memorandum will provide planning-level observations, options, schedule information, and estimated professional-service fees. Construction costs, if discussed, will be order-of-magnitude only if possible and based on the information available. A real outcome of this exercise is that construction costs may not be possible to provide until detailed assessment and options are developed at a later date.
- One consolidated round of comments on the draft memorandum is assumed. Material scope changes, additional options, repeated reviews, presentations, workshops, or meetings beyond the allowance may require additional authorization.
- Site visits, field investigations, survey, utility locating, sampling, laboratory testing, hydraulic or process modelling, equipment testing, destructive investigation, and verification of concealed conditions are excluded.
- Detailed civil, mechanical, process, electrical, environmental, structural, architectural, geotechnical, or controls design; calculations; drawings; specifications; and tender documents are excluded.
- Regulatory applications, permit submissions, agency consultation, construction procurement, contract administration, commissioning, and construction-phase services are excluded.
- Schedule performance depends on timely authorization, receipt of information, stakeholder availability, consolidated review comments, and decisions by CYNC and The City.

CLOSURE

We appreciate the opportunity to support CYNC Architecture Inc. on this project. SweetTech's objective is to provide practical and timely technical service, maintain transparent communication on effort and budget, and help the Prime Consultant define the project's needs, constraints, execution options, and recommended path forward for future assessment, design, and implementation.

Sincerely,

SWEETTECH ENGINEERING CONSULTANTS



Austyn Schutta, P.Eng., CPESC, CESSWI
Sr. Project Engineer
austyn@sweet-tech.ca
780-297-2879



EDUCATION

- 1998-2003 Bachelor of Engineering with Honors, University of Canterbury, NZ

EXPERIENCE

- Practice Lead, Mechanical and Process Engineering
  2025-Present
  SweetTech Engineering Consultants
- Senior Project Engineer
  2012-2025
  Polaris Engineering Ltd.
- Senior Project Manager
  2007-2012
  CanDyne Pump Service Inc.
- Product Design Engineer
  2003-2006
  Fisher & Paykel Appliances Ltd.

AFFILIATIONS

- Practicing Member, P.Eng., Association of Professional Engineers and Geoscientists of Alberta (APEGA)
- Practicing Member, P.Eng., Engineers and Geoscientists of Saskatchewan (APEGS)

CERTIFICATIONS

- WHMIS
- CSTS 2020
- Petroleum Safety Training (PST 2.0)
- First Aid & CPR / AED level C
- H2S Alive

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ENGINEERING CONSULTANTS

linkedin.com/in/shaunrobertson
shaun@sweet-tech.ca
www.sweet-tech.ca

SHAUN ROBERTSON, P. Eng.

Sr. PROJECT ENGINEER (MECH.)

PROFESSIONAL EXPERIENCE

Shaun has extensive experience in the design, construction, and optimization of production facilities and pipelines with over 20 years of engineering practice. He has a strong practical engineering background comprised of process design and build with significant analytical skills including extensive hydraulic analysis of various working fluid systems (Water, Oil, Gas), valve and equipment sizing, rotating equipment (pump and compression) review and sizing, process and piping system design.

Shaun has worked on various sized green and brown field projects from conception to completion requiring deep collaboration with clients to undertake project feasibility studies, define project scope, budget estimation, schedule development, project design and review through to procurement of equipment and systems, manufacturing/fabrication, construction, and commissioning. He has developed strong project management skills for complex, multi-phased projects consistently delivering on time, on budget projects.

Shaun has strong team and personnel leadership skills along with excellent coordination and management abilities. He is a proven leader in training, mentoring, motivating, and supervising both junior employees and contractors. He has a strong understanding of workplace safety, critical thinking, problem solving and the application of regulatory requirements. He also has extensive experience working with regulatory bodies such as the AER, NEB, FERC, ABSA, and TSASK.

SELECT PROJECT EXPERIENCE

Canmore Nordic Centre Upgrades, Canmore, AB (2023 – Present) – Sr. Project Engineer for the snowmaking system upgrades, including pump system review and rebuild with approximately 30 km of water and air pipelines removed and replaced in multiple phases requiring regulatory and existing facility licensing review, assessing existing equipment and pipelines, system design, procurement, construction assistance, and facility startup.

Gas Supply Pipeline, La Crete, AB, (2020 – 2023) – Sr. Project Engineer, for a 100 km sweet spec natural gas supply pipeline to Northern Canada to supply Northern communities. The project required project/personnel management including process, facility and equipment design, material selection, procurement, construction, and start up.

Oil Battery and Water/Polymer Injection Facility Upgrades, AB (2014 – 2016) – Project Engineer and Manager, staged upgrades to existing sour oil battery and water/polymer injection facilities. Shaun contributed to the multiple phases of the project from Polymer facility water supply upgrades to oil battery treater and gas gathering upgrades. Shaun was involved in the design, procurement, project/personnel management of all project phases including process, facility and equipment design, construction and start up.

International Gas Gathering Pipeline, SK (2012 – 2016) – Project Engineer and Manager, Sweet gas gathering pipeline crossing the international border from Saskatchewan, Canada to North Dakota, USA. Shaun was responsible for regulatory applications, as well as the design, procurement, and project/personnel management for the pipeline project. His duties included process, facility, and equipment design, construction oversight, and system startup.

Facility Slop Tank/Containment Upgrades, AB (2016) – Project Engineer and Manager, Shaun was involved in the facility upgrades to an existing oil battery in Alberta, Canada. Including the installation of multiple tanks, pumps, and a creative secondary containment solution. In addition to the design work, he contributed to procurement and project management, overseeing facility and equipment design, construction, and system startup.

Various Turnkey Pump Package Projects (2007 - 2012) – Project Engineer and Manager, Shaun was involved in the design, construction, and project management of multiple turnkey pump packages concurrently, with pump systems ranging from fractional horsepower to several thousand horsepower.

Additional project experience available upon request.
Visit our website www.sweet-tech.ca for more information.



EDUCATION

- 2012 B. Sc., Civil Engineering Co-op, University of Alberta

EXPERIENCE

- Civil Engineer 2016 –Present SweetTech Engineering Consultants
- Civil E.I.T. 2012-2015 SweetCroft Engineering Consultants
- Project Scheduler/Estimator 2011-2012 ATCO Electric

AFFILIATIONS

- Practicing Member, P.Eng., Association of Professional Engineers and Geoscientists of Alberta (APEGA)
- Practicing Member, P.Eng., Engineers and Geoscientists BC (EGBC)
- EnviroCert International Inc. (CPESC, CESSWI)
- Canadian Dam Association

CERTIFICATIONS

- Principles of Health and Safety Management
- Alberta OHS Legislation Awareness ATV Training
- COR Auditor
- CSTS
- First aid & CPR lvl C
- Ground Disturbance for Supervisors
- Leadership for Safety Excellence OSSA Regional Orientation

SWEETTECH
ENGINEERING CONSULTANTS

linkedin.com/in/austynschutta
austyn@sweet-tech.ca
www.sweet-tech.ca

AUSTYN SCHUTTA, P. Eng., CPESC, CESSWI

SENIOR CIVIL & HYDROTECHNICAL ENGINEER

PROFESSIONAL EXPERIENCE

Austyn is a professional civil and hydrotechnical engineer with 14 years of consulting and prime-contractor experience in multidisciplinary assessment, project management, water resources, water quality, heavy civil construction, flood mitigation, and Care of Water. He leads projects from site investigation and options development through detailed design, regulatory coordination, tendering, construction contract administration, and close-out. His practical field background supports constructable, cost-conscious solutions and effective coordination among clients, contractors, regulators, and specialist disciplines. Austyn is registered as a Professional Engineer in Alberta and British Columbia and holds Certified Professional in Erosion and Sediment Control (CPESC) and Certified Erosion, Sediment and Storm Water Inspector (CESSWI) credentials.

SELECT PROJECT EXPERIENCE

Sikome Lake / Sikome Aquatic Facility Refurbishment, Alberta Forestry and Parks, Calgary, AB, 2021 to Present – Engineer of Record and coordinating professional for the multidisciplinary assessment and phased redevelopment of an artificial lake facility. Led infrastructure and operational assessments, including water treatment, reservoir, mechanical, electrical, SCADA, environmental, geotechnical, access, grading, and cost-planning inputs.

Canmore Nordic Centre Snowmaking System Study – Phase II, Alberta Forestry and Parks, Canmore, AB, 2023 – Project lead for the condition and lifecycle assessment of an existing pumped-water system. Coordinated historical and operational reviews, field testing, pipeline investigations, specialist engineering input, options evaluation, cost estimating, and prioritized recommendations for repairs, replacement, and pumphouse upgrades (~34M).

Fording River Coal Mine Water Management System Expansions, Teck, BC – Lead engineer for Care of Water planning and environmental field programs supporting multi-phased water management infrastructure expansions exceeding $8 million. Responsibilities included hydrologic assessment, field investigations, water-quality sampling and monitoring, environmental coordination, erosion and sediment control, adaptive management, and construction support.

Grassi Lakes Redevelopment, Alberta Forestry and Parks, Canmore, AB, 2020 to 2023 – Engineer of Record and Project Engineer/Manager for a multidisciplinary infrastructure refurbishment in environmentally sensitive terrain. Led coordination from site assessment and options development through civil and stormwater treatment and design, regulatory support, tendering, construction contract administration, field reviews, submittal reviews, change management, and project close-out.

Red Deer River Flood Mitigation, Town of Drumheller, AB, 2021 to 2024 – Water resources engineer supporting assessment and upgrades to the Downtown Drumheller dike which included sub-project involving assessment and engineering of existing Spray Park water treatment and discharge realignment. Responsibilities included client and discipline coordination, conceptual and detailed design, regulatory support, cost and quantity development, tender support, construction contract administration, field reviews, and issue resolution.

Additional project experience available upon request.
Visit our website www.sweet-tech.ca for more information.

Catch Engineering
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Fax: (403) 770.8728
www.CatchEngineering.com

Scott Pederson, P.Eng.

Electrical Engineer Lead

Summary of Qualifications

Scott is a Professional Electrical Engineer, registered with APEGA and EGBC, with over 20 years of experience leading the design, coordination, and review of complex electrical systems across multiple industries, including petrochemical, conventional and heavy oil, oil sands, mining, underground residential distribution (URD), power generation, independent power production (IPP), renewable energy and wastewater treatment. Scott is responsible for technical governance, engineering assurance, and design certification, ensuring that all electrical deliverables meet project, client, and regulatory requirements from conceptual and FEED design through detailed engineering, construction, and commissioning.

Scott's expertise includes coordination and review of electrical deliverables, encompassing single-line diagrams, equipment specifications, protection and control (P&C) schematics, grounding and area classification drawings, and technical bid evaluations. He leads and participates in design reviews, 3D model validations, HAZOP and risk assessments, ensuring multi-discipline alignment and adherence to stakeholder, operations, and maintenance requirements. His experience extends to equipment procurement, vendor inspections, factory and site acceptance testing, and management of utility interconnections, electrical safety programs, and inspection compliance. Known for his methodical approach and leadership, Scott consistently delivers technically sound, standards-compliant, and constructible electrical systems for large-scale industrial and power generation projects.

Work Experience

2025 – Present | Catch Engineering | Calgary, AB

Electrical Engineer Lead

- Lead electrical QA/QC and design verification for industrial facilities, ensuring all medium- and high-voltage (MV/HV) electrical deliverables comply with CSA, IEEE, IEC, and client engineering standards.
- Review and approve engineering deliverables, QA/QC reports, and commissioning documentation, maintaining technical accuracy and traceability throughout the project lifecycle.
- Direct procurement planning, including the development and evaluation of RFQs, technical bid evaluations (TBEs), vendor selection, and contract negotiation for major equipment (transformers, switchgear, inverters, and battery enclosures).
- Perform QA/QC oversight of vendor and contractor engineering packages, validating design alignment with corporate standards, CEC, and applicable environmental classifications.
- Author and review area classification reports to confirm hazardous zone boundaries, equipment suitability, and electrical installation integrity per CEC Section 18, API RP 505, and IEC 60079.
- Verify interface consistency between electrical, instrumentation, and control drawings to ensure functional integrity across substation, process, and utility power systems.
- Review and provide feedback on internal and third-party engineering deliverables, identifying non-conformances and issuing corrective actions to maintain project compliance and traceability.
- Support commissioning and energization planning, coordinating with utility operators to manage outage windows, protection testing, and system integration activities.
- Support commissioning and field verification activities by validating as-built drawings, loop checks, and inspection test plans (ITPs) against design intent and QA/QC documentation.

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www.CatchEngineering.com

Scott Pederson, P.Eng.

2024 – 2025 Northriver Midstream Ltd Calgary, AB

Electrical Project Engineer

- Project planning and execution from FEED through detailed engineering, procurement, construction, commissioning & start-up, and operations support
- Leadership and guidance to Engineering and Procurement contractors for quality and code compliance of electrical engineering deliverables.
- Accountable for technical adequacy of electrical, instrumentation, and control system engineering activities and deliverables
- Accountable for electrical, instrumentation, and control system engineering work processes implemented.
- Ensuring electrical, instrumentation and control system design is Optimised for Safety, Reliability and Life Cycle Cost
- Support to Project Schedule, Cost Control, Change Management, and Risk Management
- Ensuring all Stakeholder input is addressed in design, including that of Operations and Maintenance
- Participation in Design, 3D Model, HAZOP, SIL, and Risk Reviews
- Participation in Life Cycle Value Assessments
- Checking engineering drawings, work packages, calculations, studies, technical bid summaries, and equipment vendor data/drawings.
- Equipment Specifications, Procurement, Inspections, and Witness Testing
- Maintaining Technical Documentation and ensuring engineering adheres to corporate DMG standards.
- Development and Maintenance of Project Electrical Standards and Specifications
- Ensure “Lessons Learned” are carried forward and applied in project electrical, instrumentation and control system design.
- Field Engineering - Technical support to Construction and Commissioning & Start-Up Teams, RFIs
- Project Coordinator for EQMP – Management of Electrical Inspection Program and CE Code Compliance
- Project Coordinator for project power supply utility service connections
- Project Coordinator for Corporate Workplace Electrical Safety Standard
- Projects included:
  - Gordondale West Gas Plant, 160mmscfd, NRM’s Largest Ever FID, $120M TIC
  - Gordondale East Gas Plant Expansion, $50M TIC
  - Dawson Creek Processing Plant Electrification $25M TIC

2017 – 2024 Inter Pipeline Ltd Calgary, AB

Senior Staff Engineer, Electrical

Projects included:

- HPC PolyPropylene Plant $1.5B, total HPC $5B, first ever combined PDH/PP facility in Canada

2014 – 2017 Jacobs Canada Inc Calgary, AB

Principal Electrical Engineer

Projects included:

- Northwest Redwater Sturgeon Refinery – first new refinery in Canada since the 80’s
- Suncor Meadow Creek DBM 120 kbpd SAGD, $1.2B
- ConocoPhillips Surmont 2 Integrated SAGD WellPads, 25 kbpd, $350M

2010 – 2014 Osum Oil Sands Corp Calgary, AB

Electrical Engineering Lead

Projects included:

- Taiga 45 kbpd SAGD $1.2B
- Sepiko Kesik 60 kpbd SAGD/CSS $2B

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www.CatchEngineering.com

Scott Pederson, P.Eng.

- Orion (Phase 2) 20 kbpd SAGD $300M

1995 – 2013 Major Technical Skills & Expertise Calgary, AB

- Kinder Morgan, TMX Mainland Terminals $3B
- Millennium Vacuum Unit Expansion $2B
- Millennium Coker Unit Expansion $2.5 B
- Millennium Naphtha Unit Expansion $2B
- U2 Debottlenecking
- Power plant electrical safety program development, safety audits, and delivering safety training to power plant staff
- PCS: Potash mines electrical safety program development, safety audits, and delivering safety training to mine staff
- TransAlta – Centralia power plant electrical safety program development, safety audits, and delivering safety training to power plant staff
- Provided lead electrical engineering and QA/QC oversight for multi-billion-dollar brownfield and greenfield oil sands and terminal expansion projects.
- Reviewed and approved electrical design deliverables, including single-line diagrams, area classifications, P&C schematics, wiring diagrams, grounding layouts, and cable schedules, ensuring compliance with CSA, IEEE, and client standards.
- Performed technical bid evaluations (TBEs) for MV/HV equipment—switchgear, transformers, MCCs, VFDs, and protection relays—verifying technical conformance and lifecycle performance.
- Led drawing and design coordination reviews, verifying consistency across electrical, instrumentation, and control systems, and ensuring constructability in congested brownfield environments.
- Conducted area classification reviews and report validation, confirming hazardous location design compliance with CEC Section 18, API 505, and IEC 60079.
- Oversaw vendor QA/QC and FAT activities for substation and field equipment, validating test data, protection settings, and functional integration.
- Authored and reviewed electrical QA/QC reports, compliance summaries, and non-conformance resolutions.
- Supported construction and commissioning by resolving RFIs, reviewing redlines, and validating as-built documentation to maintain alignment with approved electrical design intent.

Education & Certifications

1995 University of Saskatchewan Saskatoon, SK
B. Sc. Electrical Engineering

1990 St. Andrews College Laurinburg, USA
B. A. Mathematics

1988 Des Moines Area Community College Ankeny, USA
A. Sc. Physics

Professional Affiliations

- Professional Engineer (P.Eng.) of APEGA- Active Member

Catch Engineering
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Fax: (403) 770.8728
www.CatchEngineering.com

Scott Pederson, P.Eng.

- Professional Engineer (P.Eng.) of EGBC – Active Member
- IEEE/IAS/PES – Senior Member
- EIEEA member
- CSA ICWC01 member
- CSA ICWC03 member
- CSA ICWC05 member
- Alberta Safety Codes Council, Electrical Utilities Sub-Council member

Additional Training

- U of A – 21st Century Energy Transition
- Fierce Conversations Leadership Course
- First Aid Standard Class C with AED
- WHMIS and H2S Alive Certification
- Construction Safety Training System (CSTS)
- Practical Loss Control Leadership Program
- Eagle's Flight – Real Time Coaching & Front-Line Leadership Programs
- Seven Habits of Highly Effective People
- Leader's Triad
- SAIT/PMI – Project Management Professional (PMP) Training
- U. of A. - Intensive Short Course for Oil Sands Bitumen Extraction and Upgrading
- Canadian Electricity Forum - Over-current and Short Circuit Protection
- Ramco - Class I Hazardous Locations
- Canadian Electricity Forum - Electrical Protection
- GE Multilin - Concepts of Motor Protection
- Canadian Electricity Forum - Power System Protection and Control
- EPIC - High Voltage Substation Design, Maintenance and Testing
- GE - Medium Voltage Motors and Drives School
- EPIC – Mechanical Engineering for Non-Mechanical Engineers



Cory Albers

P.Eng., M. Sc.
Wetland Hydraulic Systems Engineer,
Co-Owner - Source2Source Inc.

Cory Albers is a Wetland Hydraulic Systems Engineer and Co-Owner of Source2Source Inc., specializing in naturalized, self-sustaining water treatment systems. He integrates advanced hydraulics with ecological design to create resilient, chemical-free wetlands for stormwater and treated sewage across diverse flow conditions.

A key part of Cory's work is Forensic Hydraulics—diagnosing failures and performance issues in hydraulic treatment and conveyance systems by analyzing behavior from first principles. His ability to pinpoint root causes and redesign underperforming systems has made him a valued specialist on complex water infrastructure projects.

His design philosophy emphasizes minimal energy use, no chemical inputs, resilience under all flow events, long-term operational simplicity, and integration with ecological and public landscapes.

For over 20 years, Cory has collaborated with Bernie Amell to develop industry-leading wetland and biofiltration systems, including the patented WaterKidney® and Nautilus Pond® technologies. His work spans projects at the intersection of water treatment, public space, habitat enhancement, and hydraulic performance.

EDUCATION

B.Sc. in Civil Engineering (Distinction)
University of Alberta, Edmonton, Alberta, 1994

M.Sc. in Civil & Environmental Engineering
University of Alberta, Edmonton, Alberta, 1997

CREDENTIALS

Professional Engineer, Association of Professional Engineers and Geoscientists of Alberta

Professional Engineer, Association of Professional Engineers and Geoscientists of Saskatchewan

M.Sc., U. of Alberta, Civil & Environmental Engineering, 1997

B.Sc., U. of Alberta, Civil Engineering, 1994

ACHIEVEMENTS

2024 Water Resources Award of Excellence,
Consulting Engineers of Alberta
Bonnybrook Wastewater Treatment Plant – Plant D Expansion

2021 Jury's Award of Excellence,
Canadian Society of Landscape Architects
Dale Hodges Park – Large-Scale Public Landscapes

Co-Inventor of the WaterKidney® and Nautilus Pond® systems

Lead hydraulic engineering analyst
for multiple large-scale water, wastewater, and stormwater treatment facilities

Source2source Inc. | 1601 17A Street SE Calgary, Alberta | cory.albers@source2source.ca

RELEVANT PROJECT EXPERIENCE

2023 **Dawson's Landing Pond 2B – Stormwater Design**
Chestermere, AB
Hydraulic system design for a WaterKidney® and Nautilus Pond® system within a major storm pond serving a 100 ha residential development. The system treats stormwater using naturalized processes while generating clean flows to support an adjacent natural wetland. Peak inflow: 9,000 L/s; peak treated water production: 50 L/s.

2022 **Dawson's Landing Pond 2A – Stormwater Design,** (DP)
Chestermere, AB
Hydraulic engineering for a WaterKidney® and Nautilus Pond® system serving a 90 ha residential development. Designed to reduce long-term maintenance costs while enhancing the development's ecological value. Peak inflow: 7,000 L/s; treated water production: 40 L/s.

2022 **Highfield Stormwater Pond – Integrated Biofiltration Wetland,**
Calgary, AB
Hydraulic structure design for a Nautilus Pond® system with integrated biofiltration pockets and constructed wetland habitats. Peak inflow: 20,000 L/s. Construction completed in 2022.

2023 **Livingston Pond E – Stormwater Wetland & Treatment System,**
Calgary, AB
Hydraulic engineering and structure design for a 10.5 ha stormwater management system incorporating WaterKidney® and Nautilus Pond® technologies. Peak inflow: 20,000 L/s; WaterKidney® recirculation capacity: 150 L/s.

2021 **Dale Hodges Park – Stormwater Retrofit Wetland,** (DH)
Calgary, AB
Lead hydraulic designer for a large end-of-pipe naturalized stormwater treatment system, including a Nautilus Pond®, diversion structures, and a multi-cell constructed wetland integrated into a public landscape.
The project received the 2021 CSLA Award of Excellence.

2024 **Centralized Stormwater Reservoir & Outfall,**
Chestermere, AB
Hydraulic engineering and wetland design for a WaterKidney® system serving a 1700 ha catchment draining multiple storm ponds. Peak constructed wetland flow: 100 L/s.

2014 **Bonnybrook Wastewater Treatment Plant,** Calgary, AB
Hydraulic specialist for a 1,390 ML/D treatment plant upgrade, including analyses of complex hydraulic interactions across process streams.

2013 **Vale Mine Ocean Outfall Remediation,**
New Caledonia
Forensic hydraulic analysis of energy dissipation and air-entrainment failure mechanisms affecting a deep-water ocean outfall for Vale Minerals. Designed remedial solutions to prevent recurrence.

2011 **Seymour–Capilano Filtration Plant Raw Water Supply System,**
Metro Vancouver, BC
Hydraulic analysis of two major raw water conduits (1800 ML/D total) including operational modelling for 2250 hp pump trains and gravity conveyance.

